Supplementary material

Supplementary Table 1 Literature search strategy

Database	Search strategy	Number of citations
PubMed	("PEG"[Title/Abstract] OR "percutaneous endoscopic gastrostomy"[Title/Abstract] OR ("endoscopic"[Title/Abstract] AND ("gastrostomy"[MeSH Terms] OR "gastrostomy"[Title/Abstract]))) AND ((("radiologic"[Title/Abstract] OR "fluoroscopic"[Title/Abstract] OR "radiological"[Title/Abstract]) AND ("gastrostomy"[MeSH Terms] OR "gastrostomy"[Title/Abstract])) OR ("percutaneous radiologic gastrostomy"[Title/Abstract] OR "PRG"[Title/Abstract] OR "radiologically inserted gastrostomy"[Title/Abstract] OR "RIG"[Title/Abstract])) AND ("complication*"[Title/Abstract] OR "adverse event*"[Title/Abstract] OR "adverse effect*"[Title/Abstract] OR "mortality"[Title/Abstract] OR "outcome*"[Title/Abstract] OR "mortality"[MeSH Terms] OR "mortality"[Title/Abstract])	218
EMBASE	#1. 'percutaneous endoscopic gastrostomy'/exp OR 'percutaneous endoscopic gastrostomy' OR 'percutaneous endoscopic gastrostomy':ti,ab OR 'peg':ti,ab #2. 'percutaneous radiologic gastrostomy':ti,ab OR 'prg':ti,ab OR 'radiologically inserted gastrostomy':ti,ab OR 'rig':ti,ab OR 'fluoroscopically placed gastrostomy':ti,ab OR 'fpg':ti,ab #3. 'complication'/exp OR 'complication' OR 'adverse event'/exp OR 'adverse event' OR 'adverse outcome'/exp OR 'adverse outcome' OR 'mortality'/exp OR 'mortality' OR 'complication*':ti,ab OR 'adverse event*':ti,ab OR 'adverse effect*':ti,ab #4. #1 AND #2 AND #3	187
Cochrane	#1 MeSH descriptor: [Gastrostomy] explode all trees #2 "endoscopic":ti,ab #3 #1 AND #2 #4 "percutaneous endoscopic gastrostomy":ti,ab OR "peg":ti,ab #5 #3 OR #4 #6 "radiologic":ti,ab OR "fluroscopic":ti,ab OR "radiological":ti,ab 9692 #7 #6 AND #1 #8 "percutaneous radiologic gastrostomy":ti,ab OR "prg":ti,ab OR "radiologically inserted gastrostomy":ti,ab OR "rig":ti,ab OR "fluoroscopically placed gastrostomy":ti,ab OR "fpg":ti,ab #9 #7 OR #8 #10 "complication*":ti,ab OR "adverse event*":ti,ab OR "adverse effect*":ti,ab #11 MeSH descriptor: [Mortality] explode all trees #12 #10 OR #11 #13 #5 AND #9 AND #12	9

Supplementary Table 2 Modified Newcastle-Ottawa scale (NOS) and JADAD quality scale

Study [ref.] year	Selection				Comparability	Outcome			Total score
	Question 1	Question 2	Question 3	Question 4	Question 1	Question 1	Question 2	Question 3	
Allen 2013 [15]	1	1	1	1	0	1	1	0	6
Barkmeier 1998 [46]	1	0	1	1	0	0	1	1	5
Blondet 2010 [16]	1	1	1	1	2	1	1	1	9
Cherian 2019 [17]	1	1	1	1	0	1	1	1	7
Chio 2004 [18]	1	0	1	1	1	1	1	1	7
Clayton 2019 [14]	1	1	1	1	0	1	1	1	7
Cosentini 1998 [39]	1	1	1	1	0	1	1	1	7
Desport 2005 [19]	1	0	1	1	2	1	1	1	8
Elliot 1996 [20]	1	1	1	1	0	1	1	1	7
Galaski 2009	1	1	1	1	0	1	1	1	7
Grant 2009 [21]	1	0	1	1	0	1	1	1	6
Kohli 2021 [5]	1	1	1	1	2	1	1	1	9
Kohli 2021 [1]	1	1	1	1	2	1	1	1	9
Kohli 2022 [34]	1	1	1	1	0	1	1	1	7
La Nauze 2012 [22]	1	1	1	1	2	1	1	0	8
Laasch 2003 [41]	1	1	1	1	0	1	1	1	7
Laskaratos 2013 [23]	1	1	1	1	2	1	1	1	9
Maasarani 2021 [37]	1	1	1	1	2	1	1	1	9
Maclean 2007 [45]	1	1	1	1	0	1	1	1	7
Mcallistor 2013 [24]	1	1	1	1	0	1	1	1	7
Mcdermott 2015 [25]	1	1	1	1	2	0	1	1	8
Moller 1999 [26]	1	1	1	1	0	1	1	1	7
Neeff 2003 [40]	1	1	1	1	0	0	1	1	6
Park 2019 [43]	1	1	1	1	2	0	1	1	8
Pruthi 2010 [27]	1	1	1	1	2	1	1	1	9
Righetti 2021 [44]	1	1	1	1	0	1	1	1	7
Rio 2010 [28]	1	0	1	1	0	1	1	1	6
Rustom 2006 [29]	1	1	1	1	0	1	1	1	7
Silas 2005 [30]	1	1	1	1	2	1	1	1	9
Throntron 2002 [42]	1	0	1	1	0	0	1	1	5
Vidhya 2018 [31]	1	1	1	1	2	1	1	1	9
Wollman 1997 [32]	1	1	1	1	0	0	1	1	6

JADAD quality score

	Randomization	Blinding	Withdrawals	
Singh 2017 [38]	2	2	1	5